A Hybrid Decomposition-Based Deep Learning Approach for Enhanced Ultra-Short-Term Wind Power Forecasting

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Abstract— **The massive demand for fossil fuels has negatively impacted our climate, driving increased interest in renewable energy sources such as wind, solar, and hydraulic power. Although wind turbine generators offer a promising alternative, the non-stationary nature of wind poses challenges for maximizing efficiency. To address these complexities, this study proposes a hybrid prediction model that integrates Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), Empirical Wavelet Transform (EWT), and Gated Recurrent Unit (GRU) modules. These components collectively reduce noise and stabilize the data, thereby enhancing the predictive accuracy of the GRU. The effectiveness of the CEEMDAN–EWT–GRU model is rigorously evaluated through structured experiments, each highlighting different facets of the approach. Notably, results from one of the experiments show that the proposed method improves RMSE by approximately 9.74, MAE by approximately 13.15, and R^2 approximately 0.41 compared to the benchmark model. This underscores the proposed method's superior forecasting capabilities.**

I. Introduction

As humanity witnesses rapid population growth and technological advancement, the demand for energy has increased significantly, raising concerns about sustainability and environmental impact. Nonrenewable energy sources, which contribute to greenhouse gas emissions, are limited and pose significant long-term environmental risks. As a result, there is a global shift toward renewable energy sources such as wind, solar, and hydropower. Among these sources, wind energy is gaining attention for its clean and abundant nature. For many years, researchers have been studying how technology can be used to forecast wind power and mitigate these drawbacks.

Wind power forecasting is classified into four types according to forecast horizons [1]: long-term (years), medium-term (weeks or months), short-term (up to 3 days), and ultra-short-term (up to 4 hours). The first three categories are commonly used to plan wind farms and schedule maintenance, while ultra-short-term forecasting supports the management of the power grid [2]. Various approaches have been proposed for ultra-short-term wind power forecasting, including Numerical Weather Prediction (NWP) [3], statistical models [4], and Machine Learning (ML) techniques [5]. Although recent models demonstrate high performance, there remains a notable deficiency in studies investigating the models' robust performance across different input lengths. Furthermore, few studies have explored the integration of CEEMDAN [6] and EWT [7] with deep sequential models to more effectively handle noisy wind signals. This gap highlights an opportunity to improve forecast accuracy through hybrid architectures.

This study introduces a robust hybrid framework designed to improve the accuracy of wind power forecasting. It combines the strengths of CEEMDAN, EWT, and GRU models. The primary purpose of CEEMDAN is to decompose the original data. EWT focuses on the refinement of high-frequency components, while the GRU utilizes these processed components to forecast wind power.

The key contributions of this study are summarized below.

- We propose a hybrid framework having robust performance across various input lengths for ultra-short-term wind power forecasting.
- Extensive experiments are conducted to compare the proposed model with various baseline models, including RNN, LSTM, GRU, Bi-LSTM, and combinations of these baselines with CEEMDAN and EWT.
- We examined the influence of integrating external meteorological variables into a decomposition-based model.
- A dedicated experiment has been designed to investigate the effectiveness of processing high-frequency and low-frequency time series separately.

II. Related Work

The demand for forecasting wind energy has been present for many years. Initially, methods such as Numerical Weather Prediction (NWP) and statistical approaches were extensively used, each with distinct strengths and weaknesses. NWP techniques rely on atmospheric and physical laws for forecasting. For instance, Liu et al. [3] employed the Rank Bayesian Ensemble (RBE) and a fluctuation awareness model to stabilize NWP outputs in ultra-short-term forecasting. In contrast, Chen et al. [8] and Hu et al. [9] applied Gaussian Processes (GP) to correct NWP data by incorporating spatial correlations and meteorological variables. Despite these advancements, these methods heavily depend on meteorological features, which lack historical trend analysis.

Statistical approaches, including autoregressive models and their variants, have also played a crucial role in wind power forecasting. Torres et al. [4] and Hill et al. [10] employed Autoregressive Moving Average (ARMA) and Autoregressive (AR) models for short-term forecasting, effectively capturing temporal and spatial correlations. Han et al. [11] introduced hybrid models merging ARMA with Kernel Density Estimation (KDE) to capture nonlinear dynamics. Although these methods address key physical modeling challenges of a lack of historical trend analysis, there are

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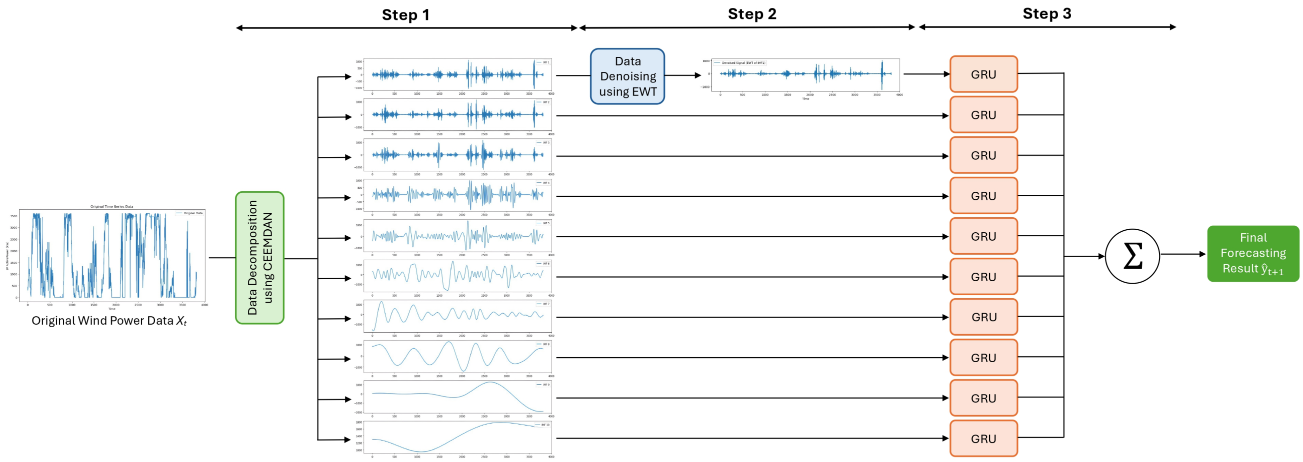


Fig. 1. Proposed CEEMDAN-EWT-GRU framework for wind power forecasting.

limitations related to nonlinearity, and volatile wind dynamics.

ML techniques have thus emerged to address complex, non-linear patterns in wind power data. Traditional ML models, such as Support Vector Machines (SVM), Random Forests (RF), k-Nearest Neighbors (KNN), and Neural Networks (NN), offer improved predictive capabilities over purely statistical methods. Jorgensen et al. [5] compared various ML algorithms and found that SVM and NN performed consistently well, but they demand careful parameter tuning and can be computationally intensive. Despite promising results, these methods still require extensive parameter optimization and reliance on feature selection.

Deep Learning (DL), a powerful subset of ML, further improves wind power forecasting by capturing sequential patterns and nonlinear dependencies. Popular architectures include Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long-Short-Term Memory (LSTM), and Gated Recurrent Units (GRUs). Yu et al. [12] used CNN to model spatio-temporal features, halving prediction errors relative to traditional time series models. Although deep learning excels at capturing complex relationships, its performance decreases with noise and non-stationarity.

Lastly, decomposition techniques such as Wavelet Decomposition (WD) and Empirical Mode Decomposition (EMD) have been increasingly used alongside deep learning to handle non-stationary wind power data. WD breaks down signals into multiple frequency components [13], capturing high- and low-frequency information, while EMD adaptively decomposes signals into Intrinsic Mode Functions (IMF). These approaches substantially improve forecast accuracy by isolating stable subsequences for individual modeling [14], but can suffer from issues such as careful parameter selection (WD) and mode mixing (EMD) [15]. Recently, advanced variants such as CEEMDAN and EWT have been proposed to further mitigate noise and improve predictive performance. Although these methods have shown considerable promise in managing noisy and non-stationary time series data, research into their integration with deep learning models remains insufficient. Particularly, their application in wind power forecasting should be further investigated. Addressing these gaps could improve the precision, reliability, and practical applicability of wind power forecasting models.

III. Proposed Method

Figure 1 illustrates the proposed framework, which involves four main steps: decomposing the data using CEEMDAN, denoising the data using EWT, forecasting with the GRU architecture and generating the final result. Each step is described in more detail below.

First, we denote the input at time t as $\mathbf{X}_t = \left[x_t^{(1)},\ x_t^{(2)},\ \ldots,\ x_t^{(N)}\right]$, where $x_t^{(n)}$ is the value of the n-th feature at time t and N is the total number of features.

A. Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN)

CEEMDAN [6] is an improved univariate decomposition technique that has been developed for the EEMD by integrating adaptive noise at each stage of the process. The CEEMDAN provides residue after each Intrinsic Mode Function (IMF) extraction to ensure better spectral separation and precise reconstruction of the original signal while reducing the computational cost. As illustrated in Fig. 1, the CEEMDAN algorithm decomposes the original wind power data into 10 Intrinsic Mode Functions (IMFs). Each IMF represents a distinct frequency component, with earlier IMFs capturing higher frequencies and later ones representing lower frequencies.

Applying this algorithm to the n-th features in $\mathbf{X}_t$, we derive the following formula:

$$x_t^{(n)} \longrightarrow \left\{\mathrm{IMF}_t^{(n,1)},\ \mathrm{IMF}_t^{(n,2)},\ \ldots,\ \mathrm{IMF}_t^{(n,K)}\right\} \quad (1)$$

where $\mathrm{IMF}_t^{(n,k)}$ is the k-th IMF of the n-th feature at time t.

B. Empirical Wavelet Transform (EWT) for High-Frequency Refinement

Although CEEMDAN effectively decomposes signals into IMFs, the first IMF typically contains high-frequency noise. EWT [7] is introduced to further reduce noise while retaining the meaningful frequency components in the first IMF. Unlike traditional wavelet transforms with fixed basis functions, EWT adaptively constructs its wavelet filters by examining the signal's own Fourier spectrum. This ensures that each frequency band is specifically tailored to the data.

Fig. 2 illustrates a comparison of the IMF1 component before and after applying the EWT denoising technique. The post-processed signal demonstrates improved smoothness while effectively retaining its primary trends and temporal structure.

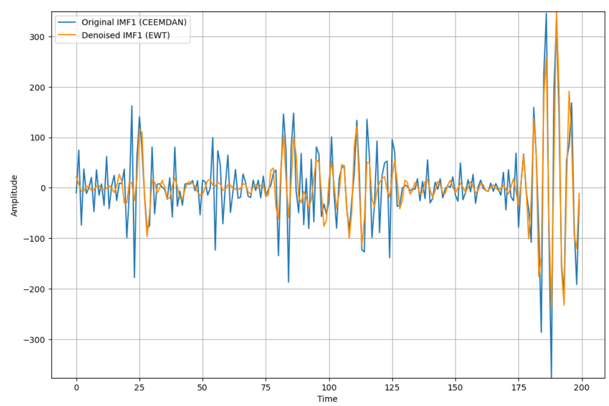


Fig. 2. Comparison of Original and Denoised IMF1.

In our proposed framework, the first IMFs of all features are denoised by EWT before being utilized to forecast future data.

$$\mathrm{IMF}_t^{(n,1)} \longrightarrow \mathrm{IMF'}_t^{(n,1)} \tag{2}$$

C. Gated Recurrent Unit (GRU)

The GRU [16] consists of two gates that are much less than the LSTM and are simple: the update and reset gates. The update gate's purpose is to control the amount of the prior gating structure retained in the current unit, while the reset gate's purpose is to determine whether the information from the past and present time steps can be closely combined. GRUs are a simplified, yet powerful variant of RNNs and LSTMs. Their basic structure provides benefits such as reducing training time without losing any notable prediction accuracy. Moreover, they are designed to capture both short- and long-term dependencies effectively while alleviating the vanishing gradient problem [17].

After decomposing and denoising the data, the k-th IMFs from all features are stacked and then fed into the GRU to forecast the k-th IMF of the wind power, which corresponds to the first feature.

$$IMF_t^{(k)} = \left[\mathrm{IMF}_t^{(1,k)}, \mathrm{IMF}_t^{(2,k)}, \ldots, \mathrm{IMF}_t^{(N,k)}\right] \tag{3}$$

For $k = 1$, replace $IMF_t^{(n,1)}$ with its EWT-transformed counterpart $\mathrm{IMF'}_t^{(n,1)}$:

$$\mathrm{IMF}_t^{(1)} = \left[\mathrm{IMF'}_t^{(1,1)},\ \mathrm{IMF'}_t^{(2,1)},\ \ldots,\ \mathrm{IMF'}_t^{(N,1)}\right] \tag{4}$$

$$\hat{\mathrm{IMF}}_{t+1}^{(1,k)} = f_{\mathrm{GRU}}^{(k)}\left(\mathrm{IMF}_t^{(k)}\right) \tag{5}$$

As shown in Figure 1, all the predicted time series are summed to generate the final forecast output. Because R_t contains mainly a noisy signal, it is eliminated from the summation.

$$\hat{y}_{t+1} = \sum_{k=1}^{K} \hat{\mathrm{IMF}}_{t+1}^{(1,k)} \tag{6}$$

IV. Experiments

A. Dataset and Input–Output Configuration

The dataset used in this study for wind power forecasting was collected in Turkey. It provides measurements of external meteorological values, and actual power output at 10-minute intervals throughout the entire year of 2018 [18]. Specifically, it contains the following features: LV Active-Power (kW), indicating the wind power output; Wind Speed (m/s), measured at turbine height; Wind Direction (degrees), describing the direction from which the wind blows at the hub height; and Theoretical Power, representing the ideal output derived from the turbine's power curve.

Each IMF produced through the decomposition of CEEMDAN-EWT was divided into training and testing sets using a split of 80:20. To structure the data for time series forecasting, a format was adopted by creating input-output pairs using a sliding-window approach. This study focuses on ultra-short-term forecasting, exploring multiple input window lengths (1 hour, 1.5 hours, 2 hours, and 4 hours), while maintaining a fixed output horizon of 10 minutes.

B. Experiment Settings

The hyperparameters used in this study include the Adam optimizer with a learning rate of 0.001, 128 units in the GRU layer, a batch size of 64, and training for 30 epochs. The loss function applied is Mean Squared Error (MSE). For implementation, the CEEMDAN decomposition is performed using the pyEMD package [19], while the Empirical Wavelet Transform (EWT) is implemented using the ewtpy package [20]. Deep learning models are built using the Keras and TensorFlow libraries. These configurations are used consistently across all experiments.

To compare the proposed method with other approaches, we employ three main evaluation metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the R-squared (R^2) score, which quantify the accuracy and reliability of wind power predictions. Following the benchmark, we ran experiments for each month and then calculated the average and sum of all metrics for every month.

Model	Metric	1 hr - 10 min		1.5 hr - 10 min		2 hr - 10 min		4 hr - 10 min	
		Avg	Sum	Avg	Sum	Avg	Sum	Avg	Sum
Transformer	RMSE	293.586	3523.036	306.499	3677.992	307.823	3693.874	300.705	3608.454
	MAE	216.004	2592.048	225.276	2703.309	241.268	2895.218	225.750	2709.004
	R^2	0.903	10.841	0.889	10.665	0.890	10.674	0.891	10.697
RNN	RMSE	322.558	3870.693	387.413	4648.958	298.447	3581.359	330.023	3960.270
	MAE	238.837	2866.045	305.683	3668.192	211.489	2537.868	253.346	3040.147
	R^2	0.879	10.542	0.824	9.891	0.893	10.717	0.870	10.436
LSTM	RMSE	355.002	4260.025	323.174	3878.088	329.386	3952.631	333.623	4003.478
	MAE	268.024	3216.290	245.355	2944.258	249.855	2998.255	257.275	3087.296
	R^2	0.865	10.378	0.881	10.573	0.877	10.521	0.873	10.477
GRU	RMSE	343.750	4124.998	311.541	3738.497	319.265	3831.182	315.186	3782.227
	MAE	270.621	3247.453	238.154	2857.844	239.535	2874.417	243.419	2921.032
	R^2	0.852	10.226	0.889	10.662	0.881	10.569	0.880	10.559
Bi-LSTM	RMSE	244.220	2930.634	239.601	2875.213	236.600	2839.204	240.979	2891.744
	MAE	152.201	1826.414	148.458	1781.491	143.160	1717.922	149.053	1788.637
	R^2	0.930	11.158	0.932	11.189	0.934	11.204	0.931	11.177
CEEMDAN-EWT-GRU (Proposed)	RMSE	**92.053**	**1104.632**	**95.485**	**1145.823**	**94.319**	**1131.832**	**93.716**	**1124.590**
	MAE	**59.831**	**717.977**	**63.540**	**762.480**	**63.108**	**757.295**	**61.491**	**737.893**
	R^2	**0.989**	**11.863**	**0.985**	**11.821**	**0.987**	**11.839**	**0.987**	**11.838**
CEEMDAN-EWT-LSTM (Benchmark)	RMSE	101.988	1223.858	97.157	1165.887	112.135	1345.617	104.615	1255.377
	MAE	68.890	826.677	65.105	781.257	77.743	932.915	71.117	853.400
	R^2	0.985	11.818	0.985	11.818	0.972	11.662	0.983	11.792
CEEMDAN-EWT-Bi-LSTM	RMSE	111.741	1340.895	101.542	1218.500	108.461	1301.536	102.127	1225.519
	MAE	77.687	932.242	68.559	822.703	75.686	908.227	68.611	823.326
	R^2	0.978	11.736	0.986	11.836	0.978	11.730	0.984	11.805
CEEMDAN-EWT-Transformer	RMSE	193.268	2319.221	197.823	2373.878	202.456	2429.474	282.123	3385.481
	MAE	133.516	1602.191	134.874	1618.486	139.113	1669.359	182.674	2192.083
	R^2	0.942	11.301	0.942	11.299	0.940	11.282	0.896	10.755

TABLE I

COMPARISON OF FORECASTING MODELS ACROSS DIFFERENT HORIZONS (FROM 1 HR–10 MIN TO 4 HR–10 MIN) USING RMSE, MAE, AND R^2 METRICS (REPORTED AS BOTH AVERAGE AND SUM).

C. Benchmarking the Proposed Model Against Baseline Models Across Various Input Lengths

This study evaluates the performance of the proposed CEEMDAN-EWT-GRU framework by comparing it with five baseline models (RNN, LSTM, GRU, Bi-LSTM, Transformer), the benchmark CEEMDAN-EWT-LSTM [21], and additional hybrid model variants as presented in Table I. The input for this experiment consists of a single feature, which is LV ActivePower.

Based on performance metrics in various historical input lengths, the proposed CEEMDAN–EWT–GRU method consistently outperforms all baseline models (RNN, LSTM, GRU, Bi-LSTM, Transformer). For the average RMSE, the value of CEEMDAN-EWT-GRU is lower than that of all baseline models by at least 140. Meanwhile, the gap in average MAE between CEEMDAN–EWT–GRU and all baseline models is at least 80. Regarding R^2, CEEMDAN–EWT–GRU achieves the highest values for all input lengths, with averages in all cases equal to or greater than 0.985. These results validate the robustness and reliability of the proposed method for ultra-short-term wind power forecasting.

In addition to baseline models, the proposed method is compared with hybrid models that integrate CEEMDAN, EWT, and baseline models, including the benchmark CEEMDAN–EWT–LSTM [21]. Across all cases, these hybrid models significantly outperform their corresponding baseline counterparts. Additionally, while Bi-LSTM and the Transformer generally exhibit superior performance compared to GRU in various input lengths, their effectiveness decreases when combined with CEEMDAN and EWT, unlike the proposed CEEMDAN–EWT–GRU. Moreover, although the benchmark model slightly underperforms relative to the proposed method, it still exceeds the performance of standalone deep learning models, underscoring the beneficial impact of incorporating decomposition techniques.

The underperformance of CEEMDAN–EWT–Transformer may be attributed to the inherent sequential nature of CEEMDAN. Although CEEMDAN addresses issues such as mode mixing and noise, it cannot be parallelized due to its iterative nature [13]. This conflicts with the Transformer's architecture, which relies on parallelized attention mechanisms and matrix operations for efficiency. Thus, CEEMDAN introduces a computational bottleneck that limits the advantages of Transformer-based models in this hybrid setting.

On the other hand, the CEEMDAN–EWT–GRU model outperforms other hybrid combinations, including CEEMDAN–EWT–BiLSTM and CEEMDAN–EWT–LSTM, aligning with the findings reported by Zhao et al. [17]. The superior performance of GRU can be attributed to its simpler architecture, consisting of only two gates (update and reset), which enables it to efficiently process decomposed signals. Compared to LSTM and Bi-LSTM, GRU requires fewer parameters and offers faster training, which is particularly advantageous after decomposition, where the burden of learning long-term dependencies has already been reduced. In this context, the additional complexity of LSTM-based models becomes redundant.

Moreover, Zhao et al. [17] also highlight GRU's superior capability in handling high-frequency components, which is especially relevant when processing IMFs derived from CEEMDAN. The reset gate in GRU allows it to dynamically disregard irrelevant historical information, making it more adaptive to localized temporal variations in high-frequency IMFs. Conversely, LSTM and Bi-LSTM models are designed to retain long-term memory, which can lead to overfitting or unnecessary redundancy when applied to decomposed signals, particularly those dominated by ultra-short-term fluctuations.

In summary, the CEEMDAN–EWT–GRU model achieves the best trade-off between accuracy and efficiency, making it highly suitable for ultra-short-term wind power forecasting involving decomposed, noise-reduced signals.

D. Advanced Modeling Strategies: Feature Enrichment and Hybrid Deep Learning with Frequency Decomposition

There are two parts to this experiment. The first part evaluates the performance of the CEEMDAN–EWT–GRU framework when additional meteorological features are included, such as wind speed, wind direction and theoretical power curve. Based on the results in Table I, a historical input of 1 hour and a forecasting horizon of 10 minutes were chosen due to the superior performance observed in RMSE, MAE, and R^2.

The second part explores a framework inspired by [17], where IMFs are categorized into high- and low-frequency components using permutation entropy (PE) [22], and then processed by different DL models to assess forecast accuracy improvement. For more details on this framework, the original data initially undergo CEEMDAN decomposition, followed by the application of EWT to the first IMF. PE values are then calculated for the resulting IMFs, which are divided into high-frequency and low-frequency sets based on a fixed threshold of 0.45 in this study. The high-frequency IMFs are input into a GRU model due to their effectiveness in capturing fast-changing patterns. In contrast, the low-frequency IMFs are processed using a Bi-LSTM, which has demonstrated the best performance among the five baseline models.

In Table II, CEEMDAN-EWT-GRU-4 includes all four characteristics of the data set. CEEMDAN-EWT-GRU-3 incorporates wind power, wind speed, and the theoretical power curve. CEEMDAN-EWT-GRU-2 and CEEMDAN-EWT-GRU-MHA-2 both utilize wind power and the theoretical power curve as inputs. Meanwhile, CEEMDAN-EWT-GRU (Proposed) and CEEMDAN-EWT-GRU-BiLSTM use only wind power as input feature.

Model	Metric	Avg	Sum
CEEMDAN-EWT-GRU (Proposed)	RMSE	**92.053**	**1104.632**
	MAE	**59.831**	**717.977**
	R^2	**0.989**	**11.863**
CEEMDAN-EWT-GRU-4	RMSE	956.152	11473.830
	MAE	807.697	9692.359
	R^2	-0.114	-1.370
CEEMDAN-EWT-GRU-3	RMSE	971.194	11654.330
	MAE	823.032	9876.389
	R^2	-0.110	-1.319
CEEMDAN-EWT-GRU-2	RMSE	140.407	1684.880
	MAE	85.736	1028.834
	R^2	0.972	11.669
CEEMDAN-EWT-GRU-MHA-2	RMSE	1046.648	12559.780
	MAE	878.553	10542.640
	R^2	-0.276	-3.308
CEEMDAN-EWT-GRU-BiLSTM	RMSE	99.222	1190.664
	MAE	66.146	793.752
	R^2	0.988	11.85

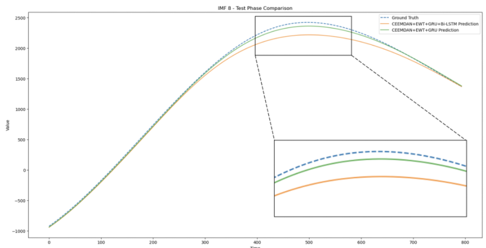
TABLE II

FORECASTING PERFORMANCE OF DIFFERENT METHODS WITH 1-HOUR INPUT LENGTH AND 10-MINUTE HORIZON

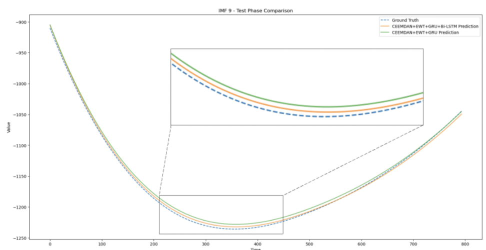
Based on the results in Table II, the single-feature CEEMDAN–EWT–GRU model performed best, producing the lowest RMSE (92.053) and MAE (59.831), alongside a high R^2 (0.989). In contrast, adding more features diminished its forecasting capabilities. Notably, using 4 features resulted in the unfavorable RMSE (956.15) and a negative R^2 value (-0.114). Reducing to three features still yielded a high RMSE (971.19) and R^2 of -0.11. Similarly, introducing Multi-Head Attention (MHA) worsened the outcome, increasing RMSE to 1046.65, MAE to 878.55, and reducing R^2 to -0.276. These findings suggest that decomposing each feature independently via CEEMDAN leads to mismatched IMFs, creating frequency-mixing issues that degrade the GRU's ability to learn meaningful patterns. This observation is consistent with the suboptimal multi-time-series decompositions reported by Yang et al. [23]. Interestingly, the two-feature model (wind power and theoretical power curve) provided more stable performance (RMSE=140.41, MAE=85.73, R^2=0.972), likely because both features represent power (kW) and have relatively aligned frequency components. Nevertheless, even this setup did not outperform the single-feature CEEMDAN–EWT–GRU, indicating that CEEMDAN's univariate decomposition principle remains incompatible with multiple unaligned features.

Furthermore, the results in Table II also shows that the proposed CEEMDAN-EWT-GRU method surpasses the hybrid method by achieving lower RMSE and MAE values. In Fig. 3, analysis of a sample shows that GRU captured trends more effectively than BiLSTM in IMF 8, which is classified as a low-frequency component. Although BiL-

STM slightly outperformed GRU in IMF 9, the difference was minimal. When aggregating all IMF predictions, the CEEMDAN+EWT+GRU model achieved better overall performance. Therefore, using BiLSTM for low-frequency IMFs did not enhance performance, while the GRU-based approach proved more efficient.



(a) IMF 8



(b) IMF 9

Fig. 3. IMF-Level Comparison Between Predictions and Ground Truth. The green line represents CEEMDAN+EWT+GRU, the orange line represents CEEMDAN+EWT+GRU+BiLSTM, and the dashed blue line indicates the ground truth.

V. CONCLUSIONS

The proposed framework combines CEEMDAN, EWT, and GRU to enhance ultra-short-term wind power forecasting. The decomposition capabilities of CEEMDAN and the denoising efficacy of EWT, when used together with baseline models, significantly improve performance. In some instances, RMSE and MAE values are reduced to a third of those observed with baseline models operating independently. Extensive experiments demonstrate that the CEEMDAN-EWT-GRU combination consistently outperforms all metrics across various historical input lengths. The framework is effective even with a single feature input. While there are experiments conducted to explore the integration of additional features or the differential processing of low- and high-frequency IMFs to boost performance further, the CEEMDAN-EWT-GRU combination remains superior. It is thus reasonable to consider this framework a robust and reliable option for ultra-short-term wind power forecasting. Future work will explore advanced decomposition techniques and frequency aligning algorithms to better integrate multiple features into the framework, potentially enhancing prediction accuracy.

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